

Data Flow Diagram Template

Date	15 March 2026
Team ID	Batch 2023-27 ACOE
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	2 Marks

Level 0 — Context Diagram

Element	Description
External Entity	Disaster Management Officer / Meteorologist
System	Rising Waters Flood Prediction System (Flask + XGBoost)
Input Flow	annual_rainfall, seasonal_rainfall, cloud_visibility, humidity, temperature, river_level
Output Flow	Binary prediction label (Flood / No Flood) + probability score + emergency protocol

Level 1 — Internal Data Flow

Proc ess ID	Process Name	Input	Output	Data Store
P1	Form Input Capture	User web form values	Raw feature dictionary	—
P2	Input Validation	Raw feature dict	Validated dict / Error msg	—
P3	Feature Scaling	Validated features	Scaled NumPy array	scaler.save

Proc ess ID	Process Name	Input	Output	Data Store
P4	ML Prediction	Scaled NumPy array	Label (0/1) + Probability	floods.save
P5	Result Routing	Prediction label	HTTP redirect to result page	—
P6	Prediction Logging	All inputs + result	New row appended	predictions_log.csv